

Smart Attendance System - Codebase Report

Overview:

- Project root: D:\ML\attendance_system

Dependencies (from requirements.txt):

- opencv-contrib-python>=4.8.0
- numpy>=1.24
- openpyxl>=3.1
- Pillow>=10.0
- ultralytics>=8.1
- mediapipe>=0.10.9
- matplotlib>=3.7

Modules:

File: attendance_logger.py

Docstring: Attendance Logger Module

Classes:

- AttendanceLogger: Manages the single master attendance workbook.
 - * `__init__`:
 - * `_create_new_workbook`: Create the master workbook with styled headers.
 - * `_date_str`: Always return today's date string - never stale, even across midnight.
 - * `_check_date_rollover`: If the calendar date has changed since the last mark, clear the
 - * `_load_todays_marks`: Scan the workbook and populate `marked_names` with students
 - * `mark_present`: Record `*name*` as present for today (no duplicates within the same day).
 - * `is_already_marked`: Return True if this student has already been marked present today.
 - * `get_present_list`: Return students marked present in the current session (today).
 - * `_flush_to_sheet`: Write all in-memory present records to the worksheet.
 - * `_autosave_loop`: Background daemon: flush + persist the workbook every `interval` seconds.
 - * `get_summary`: Return a summary dict for the current session.
 - * `save`: Flush in-memory records to the sheet and persist to disk.
 - * `close`: Flush, save, and stop the auto-save thread.

Functions:

- `_thin_border`:
- `_header_style`:
- `_data_style`:
- `get_student_history`: Read the single master attendance workbook and compute per-student stats.

File: attendance_popup.py

Docstring: Attendance History Popup

Functions:

- `show_attendance_popup`: Spawn the attendance-history popup for a student.
- `_render_pie_chart`: Render a donut-style pie chart using matplotlib and return
- `_create_popup`: Build and display the popup Toplevel window.

File: config.py

Docstring: Configuration for the Attendance Management System.

File: dashboard.py

Docstring: Smart Attendance System - Streamlit Admin Dashboard

Functions:

- `_delete_attendance_records`: Delete attendance rows from the Excel file for the given date and
- `_rename_student`: Rename a student in:
 - `_delete_student_dashboard`: Delete a student's face data and (optionally) purge their attendance rows
- `load_attendance_df`: Load the master attendance Excel into a tidy DataFrame.
- `load_enrolled_students`: Return list of {name, roll_no} from the label map pickle.

- get_student_stats: Compute per-student attendance percentage across all session dates.
- pct_color:

File: face_detector.py

Docstring: Face Detector Module

Classes:

- FaceDetector: YOLOv8-face based face detector - CPU-optimised.
 - * __init__: Initialize the face detector, downloading model if needed.
 - * _ensure_model: Download model file if it doesn't exist.
 - * detect: Detect faces in a BGR frame.
 - * detect_with_confidence: Detect faces with confidence scores.

File: face_encoder.py

Docstring: Face Encoder Module (YOLOv8-face + LBPH version)

Functions:

- _sanitize_path_component: Strip characters unsafe for filesystem directory names.
- _augment_face: Generate augmented variants of a grayscale face crop.
- enroll_from_gui: Detect a single face in a BGR frame for enrollment.
- save_enrollment: Save enrollment photos and retrain the LBPH model.
- _train_lbph_model: Train/retrain the LBPH model from all saved face images.
- _load_label_map: Load the name -> label_id mapping from disk.
- _save_label_map: Save the label map to disk.
- load_all_encodings: Get enrolled student info (compatible API).
- delete_student: Remove a student's data and retrain the model.
- get_enrolled_students: Get a list of all enrolled students.

File: face_recognizer.py

Docstring: Face Recognizer Module (OpenCV LBPH version)

Classes:

- FaceRecognizer: Handles real-time face recognition from camera frames.
 - * __init__: Load the LBPH model and label map. MediaPipe is loaded lazily.
 - * reload_model: Reload the LBPH model and label map from disk.
 - * _init_landmarker: Download (if needed) and init the FaceLandmarker - runs in BG thread.
 - * recognize_frame: Detect and recognise faces in a single frame.
 - * _draw_corner_brackets: Futuristic corner-bracket bounding box.
 - * _draw_scan_line: Animated horizontal scan line sweeping through the face bbox.
 - * _draw_confidence_bar: Thin confidence bar above the face box.
 - * _draw_hud_label: Semi-transparent HUD label below the face box.
 - * _draw_face_mesh: Draw face contour wireframe (called at most every

FACE_MESH_EVERY_N_FRAMES

- * _draw_fps: Overlay the live FPS reading in the top-right corner.
- * draw_results: Draw futuristic HUD-style bounding boxes + labels on the frame.
- * num_enrolled: Number of enrolled students.

File: main.py

Docstring: Smart Attendance Management System

Classes:

- AttendanceApp: Main application class for the Attendance Management System.
 - * __init__:
 - * _setup_styles: Configure ttk styles for a dark, modern theme.
 - * _build_header:
 - * _build_main_area:
 - * _build_default_side_panel:
 - * _build_enrollment_panel:
 - * _build_attendance_panel:
 - * _build_status_bar:
 - * _cancel_feed_loop: Cancel any pending _update_camera_feed after-callback.

- * `_start_camera`: Open the webcam and clear stale cached results.
- * `_start_worker`: Launch the background inference + render worker thread.
- * `_stop_worker`: Signal and join the worker thread.
- * `_worker_attendance`: Background: YOLO + LBPH + throttled MediaPipe mesh + PIL render.
- * `_worker_enroll`: Background: YOLO detection + scan-line annotation + FPS + PIL render.
- * `_frame_to_imgtk`: Convert BGR OpenCV frame -> PIL Image -> ImageTk.PhotoImage.
- * `_resize_with_aspect_ratio`:
- * `_stop_camera`: Close the webcam, stop worker, and reset to default state.
- * `_update_camera_feed`: Main-thread loop - only grabs raw frames and pushes to worker;
- * `_start_enrollment_mode`:
- * `_capture_enrollment_photo`:
- * `_update_enrollment_progress`:
- * `_save_enrollment`:
- * `_safe_set_enroll_status`: Update enroll_status label safely (may not exist if cancelled).
- * `_on_enrollment_saved`:
- * `_start_attendance_mode`:
- * `_on_models_loaded`:
- * `_update_attendance_list`:
- * `_stop_attendance`:
- * `_refresh_student_list`:
- * `_retrain_model`: Re-train the LBPH model and hot-reload it in the running session.
- * `_silent_retrain`: Silently retrain the LBPH model in a background thread.
- * `_delete_selected_student`:
- * `_update_enrolled_count`:
- * `_set_status`:
- * `_on_close`:

Functions:

- `main`: